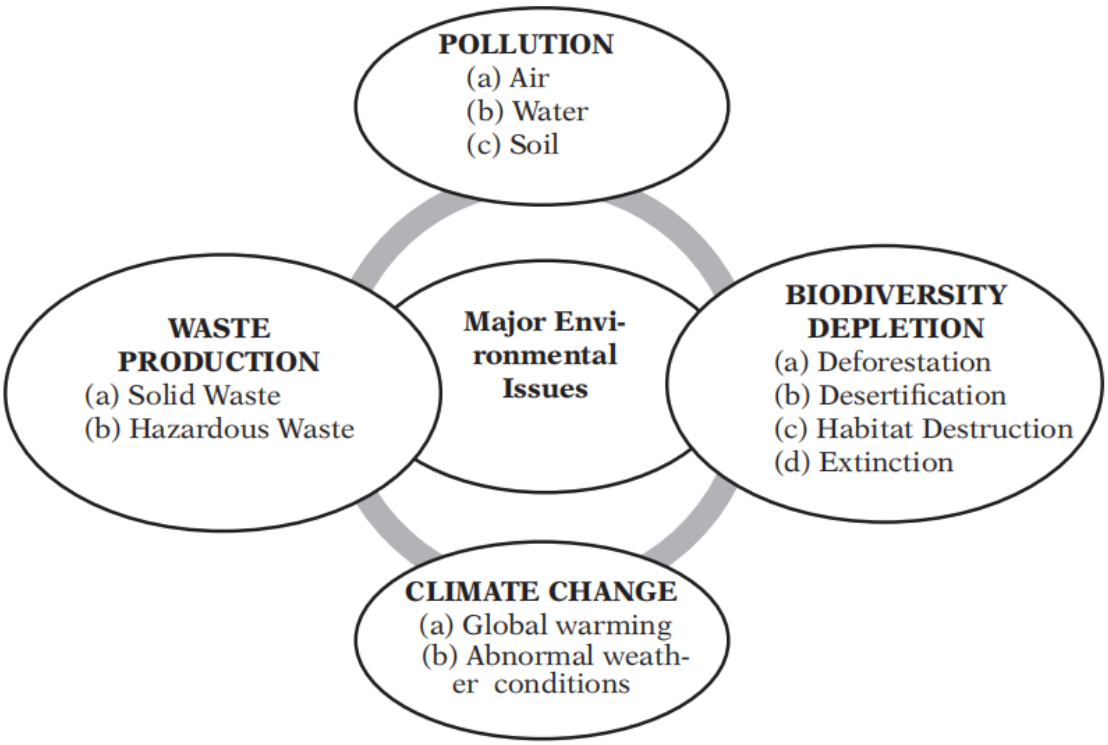
UNIT-1

Chapter-1 Introduction to Environmental Science

The term 'Environment' has been derived from the French word "Environ" which means to encircle or surround. Environment literally means the surroundings in which we are living. Environment means all living components like plants and animals and non-living components like water, air etc. surrounding us. It includes all those things on which we are directly or indirectly dependent for our survival. Environmental Protection Act (1986) defined,

"Environment as the sum total of water, air and land, their inter-relationship among themselves and with the human beings, other living beings and property."

Environmental studies envisage study of interactions of the physical, chemical and biological components of the environment and also the relationships and effects of these components with the organisms in the environment. Environmental sciences apply interdisciplinary approach for understanding the different aspects of environment and to address the environmental issues and problems. A brief idea of environmental issues and their interlinked nature is depicted below.



Fundamentally, all environmental problems involve either depletion of resources (consumption) or pollution of sinks (production of waste) as seen from the above figure. Therefore, study of these aspects and finding a sustainable solution to these problems comes under the purview of environmental studies.

The field of environmental science study can be divided into three main areas:

1. To learn how the natural world works.
2. To understand how we as humans interact with the environment.
3. To determine how we affect the environment.

Nature and Characteristics of Environment

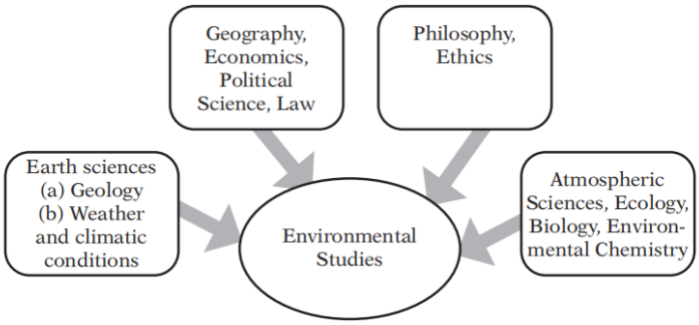
The environment is made up of living and non-living things, and it has many characteristics, including:

- **Atmosphere**: The layer of gases that contains oxygen and nitrogen, and protects life from the sun's harmful ultraviolet rays
- **Water**: A vital resource for life, found in rivers and oceans
- **Plants:** A source of oxygen and an important part of the landscape, often used for recreation
- **Animals**: An important part of the environment, and sometimes a source of tourism
- **Climate**: The long-term weather pattern in an area, which affects the types of plants and animals that live there, and the availability of water
- **Land:** The shape of the land, or landforms, can affect how people use the land
- **Natural resources**: Resources like water, metals, forests, and soils, which should be used responsibly
- **Biosphere**: The layer where life exists, including ecosystems in the air, on land, and in water

The natural environment is essential for human health and well-being, but it is impacted by human activities, such as pollution, urbanization, and climate change.

Multi-disciplinary Nature of Environmental Studies

Since, the environment is complex and actually made up of many different environments, including natural, constructed and cultural environments, environmental studies is the interdisciplinary examination of how biology, geology, politics, policy, studies, law, religion, engineering, chemistry and economics combine to inform the consideration of humanity's effects on the natural world.



It is essentially a multi-disciplinary approach and its components include Biology, Geology, Chemistry, Physics, Engineering, Sociology, Medical Sciences, Anthropology, Economics, Statistics, Philosophy etc. Role of different branches can be summarized as:

- **Physics:** To understand the change of material and energy interaction and to make mathematical models of environment.
- **Chemistry:** To understand the molecular interactions in the system.
- **Biology:** To describe the effects within the plant and animal kingdom and their diversity.
- **Atmospheric Science:** To examine the phenomenology of the Earth's gaseous outer layer. It comprises meteorological studies, greenhouse effect, airborne contaminants, sound propagation phenomena related to noise pollution, and even, light pollution.
- **Ecology:** To analyse the dynamics among an interrelated set of populations, or a population and some aspects of its environment. These studies could be on endangered species, predator interactions, effects upon populations by environmental contaminants etc.
- **Environmental Chemistry:** To study the chemical alterations in the environment. Principal areas of study include soil contamination and water pollution. The topics of analysis involve chemical degradation in the environment, multi-phase transport of chemicals and chemical effects upon biota.
- **Geo-science:** It includes environmental geology, environmental soil science, volcanic phenomena and evolution of the earth's crust.

- **Mathematics and Computer Science:** Used in environmental modelling and analysis of environment related data.
- **Economics:** It deals with economical aspects of various components of environment.
- **Law:** It helps in framing of environment related laws, acts, rules and their monitoring.
- **Social Science:** It helps in dealing with population and health related issues.

Components of Environment

Environment may be regarded as a 'space' or a 'field' in which networks of interactions between different entities occurs. A detailed discussion of components of environment is given below:

The four major components of the environment are lithosphere (rocks), hydrosphere (water), atmosphere (air), and biosphere (life):

1.Lithosphere

The term 'Lithosphere' is derived by combining the Greek word "lithos," which signifies rock, with the word "sphere." The lithosphere is the solid outer portion of

Earth which includes Earth's crust as well as the underlying cool, dense, rigid uppermost portion of the upper mantle. At the center of the Earth is a structure called the Core, which has a diameter of around 7000 km and a radius of 3500 km. Surrounding the core is the 2900 km thick mantle. The lithosphere extends from the surface of Earth to a depth of approximately 100 km. The tectonic plates of Earth make up the majority of the lithosphere.

2.Hydrosphere

It includes all types of water bodies on Earth, such as freshwater (rivers, lakes, ponds, streams), marine (oceans, seas), and groundwater resources, among others. There are three states of water: gas (water vapour), liquid (water), and solid (ice). It occupies 71% of the planet's surface. The oceans contain 97.5% of all the water on Earth, which is salt water. Freshwater makes up only 2.5% of all water on Earth. Of total, 68.9% is in frozen forms, such as glaciers, while 30.8% is accessible as groundwater. A readily available quantity of 0.3% can be found in lakes, reservoirs, and rivers.

3.Atmosphere

The earth is enveloped by thick gaseous layers that make up the atmosphere. It stretches up to 300 km. Life is supported by the atmosphere which contains a lot of oxygen. It is composed of gases such as nitrogen (78.08%), oxygen (20.95%), argon (0.93%), carbon dioxide (0.039%), and trace amounts of hydrogen, helium, and noble gases. The troposphere, stratosphere, mesosphere, thermosphere, and exosphere are the layers that rise above the surface of the earth.

- The lowest layer is called the **troposphere**. It rises from ground level to a height of roughly 11 kms above sea level.
- The **stratosphere** is the next layer above. It stretches from the top of the troposphere to about 52 km above the ground. The ozone layer is present in stratosphere. Ozone molecules in this layer absorb ultraviolet light from the sun and convert the UV energy into heat.
- The **mesosphere** is located above the stratosphere. It rises to a height of approximately 85 kms above Earth. The mesosphere is where most meteors burn up.
- The **thermosphere** is the layer of extremely rare air above the mesosphere. The thermosphere absorbs UV and high energy X-rays from the Sun, which can cause it to heat up to thousands or even hundreds of degrees. The top thermosphere can have temperatures as high as 2,000°C.

4.Biosphere

It includes every area of Earth where life is present. It includes both living things and non-living objects in the environment. There could be ecosystems in the soil, air, water, or on land that sustain life. Geologist Edward Suess first used the word "biosphere" to refer to any area of Earth where life is present. It stretches from the polar ice caps to the equator, with a living form appropriate for the local environment existing in each zone. An ecosystem is a part of the biosphere that influences the state of the planet as a whole.

Biotic and Abiotic Components of Environment

The biotic and abiotic components of the environment begin to have significance when they start interacting. For example, biotic components such as plants provide nourishment for other living organisms. The soil is an abiotic component that provides nutrients and other essential components to promote plant growth. Both biotic and abiotic factors create and rely on one another, including soil, nutrients, and other abiotic factors.

The components of the environment can be broadly divided into two categories: Biotic Components and Abiotic Components

Biotic Component of Environment

Also known as biological component of environment. It consists of flora and fauna as well as human beings that make up the ecosystem. Ecosystems are created by the interactions of plants, animals, and microorganisms with abiotic materials. These ecosystems also feature distinct populations of producers, consumers, and decomposers among other organism types. Fungi, bacteria, animals, plants, and animals are biotic factors. And based on the energy source, these biotic components can be further separated into producers, consumers, and decomposers.

Producers (Autotrophs)

These are plants and other photosynthetic organisms that use carbon dioxide, water, and sunlight to produce food on their own. They serve as the foundation of the food chain because they transform solar energy into a form that other organisms may use.

Consumers (Heterotrophs)

Consumers are the organisms which depend on producers for energy. Examples of consumers include herbivores (plant eaters), carnivores (meat eaters), and omnivores (both plant and animal eaters).

Decomposers (Saprophytes)

They decompose organic debris and dead organisms into simpler chemicals, which they then consume as food. They are crucial to the recycling of nutrients within ecosystems because they return nutrients to the soil for plant uptake.

Abiotic Component of Environment

Also known as physical component of environment. It is the non-living portion of the ecosystem. The abiotic component of an ecosystem is made up of the environment, water, light, air, soil, humidity, nutrients, and other physical and chemical factors. In general, abiotic components of ecosystem differ from one ecosystem to the other. The aquatic environment has abiotic components such as salinity, nutrients, and dissolved oxygen. Type of soil, rain, wind, temperature, altitude, sunlight, and nutrients are all significant abiotic components in terrestrial ecosystems. These abiotic components can be grouped into different categories:

- **Physical Factors:** These consist of factors including air pressure, temperature, humidity, and light intensity. These elements define an area's climate and have a big impact on the kinds of species that thrive there.
- **Chemical Factors:** These include the water and soil composition, which includes the pH, nutrition levels, and presence of different minerals. These elements have an impact on the nutrients that are available to plants and other living organisms.
- **Geological Factors:** An area's underlying geological features, including the kinds of rocks and soil, drainage, the availability of nutrients, and the general suitability of the habitat, can all be impacted.
- **Topographic Factors:** The topography of the area, including its mountains, valleys, and water bodies, affects the micro-habitats, water flow, and local weather patterns.

Scope of Environmental Studies

The field of environmental studies encompasses a wide range of scopes, which are not only important for children but also for individuals of all ages. These scopes can be summarized as follows:

1.Resource Awareness

Environmental studies foster awareness about both renewable and nonrenewable resources in a particular region. This involves assessing the available resources, their utilisation patterns and the need to maintain a balance for future generations.

2.Ecological Understanding

It imparts knowledge about ecological systems and the cause-and-effect relationships within them.

3.Biodiversity Awareness

Environmental studies provide essential information about the richness of biodiversity and the potential threats to plant, animal and microbial species in the environment.

4.Disaster Preparedness

This field helps individuals understand the causes and consequences of natural and human-induced disasters, such as floods, earthquakes, landslides, cyclones and various types of pollution. It also explores measures to minimise their effects.

5.Decision-Making

Environmental studies equip individuals with the ability to evaluate alternative responses to environmental issues before deciding on a course of action.

6.Environmental Literacy

It enables individuals to become environmentally literate citizens by providing knowledge of environmental laws, rights, rules and regulations. This knowledge empowers them to make informed judgments and decisions for the protection and improvement of the Earth.

7.Social Issues

Environmental studies shed light on societal problems like overpopulation, public health and hygiene. It also emphasises the role of arts, science and technology in addressing these challenges.

8.Eco-Friendly Technologies

The field identifies and promotes the development of appropriate, indigenous and eco-friendly skills and technologies to address various environmental issues.

9.Sustainable Resource Use

It educates citizens about the importance of sustainable resource utilisation, emphasising that these resources are inherited from our ancestors and must be passed on to future generations without compromising their quality.

10.Practical Application

Environmental studies bridge the gap between theoretical knowledge and practical application, demonstrating the multiple uses and benefits of a well-managed environment.

Importance of Environmental Science.

Numerous environmental issues have grown in scale, progressively complicating the global system and endangering humanity's existence on Earth. The significance of environmental studies can be attributed to the following reasons:

1.Global Nature of Environmental Issues

It is widely acknowledged that environmental problems, such as global warming, ozone depletion, acid rain, marine pollution and biodiversity loss, transcend national boundaries. Consequently, addressing these issues requires international collaboration and concerted efforts.

2,Development and Its Environmental Impact

The development brings about urbanization, industrial expansion, advanced telecommunications and transportation systems, high-tech agriculture and housing, among other advancements. However, it has often come at the cost of environmental degradation.

Some developed regions have shifted polluting industries to developing nations, contributing to environmental challenges. The environmental consequences of development are not limited to specific regions but affect the entire world.

3.Escalating Pollution Levels

The world's population is on the rise and India, for instance, is home to a significant portion of the global population. This puts immense pressure on natural resources, including land. Soil health problems, such as nutrient deficiencies, soil salinity and degradation, have been recognized by agricultural experts.

4.The Need for Alternative Solutions

Particularly in developing countries, there is a pressing need to chart alternative pathways towards sustainable development. This involves setting goals for development that are environmentally responsible and sustainable. Such a goal should be universal, transcending the division between the developing and over-consumptive developed world.

5.Preserving Humanity

It is of utmost importance to prevent the potential extinction of humanity due to activities that harm the environment in the name of development. Striking a balance between development and environmental protection is essential.

6.Wise Development Planning

Survival and sustenance rely on the availability of resources. Therefore, any development plan should be ecologically sound and synchronised with the natural cycles of the environment. Planning should prioritise environmental sustainability alongside economic development.

Why You Should Study Environmental Studies?

There are many reasons why you should study environmental studies, but here are 5 of the most important and compelling ones:

1. **To enhance your environmental knowledge and awareness**: Environmental studies will provide zou with the scientific knowledge and tools to measure, monitor, and model the environmental systems and processes, and to assess and mitigate the environmental impacts and risks of human activities and systems.
2. **To develop your environmental skills and competencies**: Environmental studies will equip you with the skills and competencies to analyze and solve environmental problems using scientific methods and ethical principles.
3. **To foster your environmental values and attitudes**: Environmental studies will cultivate your values and attitudes for appreciating and respecting the environment and its diversity and integrity, and for caring and acting for the environment and its protection and conservation.
4. **To empower your environmental participation and engagement**: Environmental studies will empower you to participate and engage in the environmental decision-making and governance processes, and to influence and shape the environmental policies and actions at different levels and sectors.
5. **To enable your environmental innovation and creativity:** Environmental studies will enable you to discover and develop new and sustainable technologies and practices that can enhance the efficiency and productivity of environmental resources and services, and reduce the environmental footprint of human activities and systems.

What are the main types of environmental issues/Challenges?

Each environmental issue contributes to a host of interconnected challenges facing the Earth and human beings. Here's a closer look at some of the main types of environmental issues:

1. Climate change
2. Loss of biodiversity
3. Air pollution
4. Ocean health

5. Water pollution
6. Overpopulation
7. Energy use
8. Weather events

1.Climate change

Climate change refers to the long-term changes in temperature, precipitation, and other weather patterns caused by human activities such as burning fossil fuels. These activities have increased the number of greenhouse gas emissions released; these emissions trap heat within Earth's atmosphere, leading to increasing global temperatures.

According to NASA, the Earth's average surface temperature has risen by about 1°C since the late 19th century.[1] The results include melting glaciers, rising sea levels, disrupted ecosystems and an increase in severe weather events such as droughts, floods, heat waves and wildfires.

2.Loss of biodiversity

Biodiversity refers to the variety of life on Earth, including animals, plants, and microorganisms. From the Amazon to the tundra, biodiversity is essential to the ecological balance of the planet. A loss of biodiversity can lead to species extinction, put food and water supplies at risk and reduce carbon sequestration (the natural process of removing carbon dioxide from the atmosphere, which is essential to reducing climate change).

Human activities, such as deforestation, agricultural expansion, land use changes and pollution, contribute to the overall loss of biodiversity. The use of pesticides can also harm nontarget species and disrupt ecosystems. According to the World Wildlife Fund, the Earth has lost 69% of its wildlife populations since 1970.[2]

3.Air pollution

Air pollution refers to the presence of harmful substances such as carbon dioxide, methane, and nitrogen dioxide in the air that people breathe. The burning of fossil fuels, industrial processes, transportation, and wildfires can have a negative impact on air quality. Exposure to fine particles, ground-level ozone, and other pollutants can cause respiratory problems, heart disease, cancer, and other health conditions. According to the World Health Organization, outdoor air pollution causes 4.2 million premature deaths every year.[3]

4.Ocean health

The Earth's oceans face several threats. Oceans absorb almost a third of the carbon dioxide that is released into the atmosphere; as global carbon emissions rise, so does the amount that is absorbed by oceans, leading to acidification. Ocean acidification can harm marine life, disrupt ecosystems, and impact global food security.

Pollution also puts the oceans at risk: the United Nations estimates that 11 million metric tons of plastic enter the water each year.[4] While wastewater, oil spills, chemicals, and other pollutants cause harm to living organisms and their habitats. Also, as global warming melts Earth's glaciers, sea levels rise, which can harm marine life and cause coastal flooding and erosion.

5.Water pollution

Beyond the ocean, Earth's other water supplies are also facing challenges. Safe drinking water is critical for human health; however, industrial waste, pesticides, and agricultural processes can pollute water sources. The presence of the resulting bacteria and chemical concentrations in drinking water can cause digestive problems, neurological illnesses, skin infections and more.

More than a billion people worldwide do not have access to clean water. As climate change and human actions shrink the available water supply, two-thirds of the world's population may face water shortages by 2025.[5]

6.Overpopulation

According to the United Nations, the world's population is expected to reach 9.7 billion by 2050.[6] As the global population grows, so does demand for natural resources, as well as human impact on the environment. Without sustainable development, overpopulation can lead to shortages in food and water and other resource depletion. It can also exacerbate issues like waste disposal, pollution, and deforestation that can contribute to public health problems.

7.Energy use

Overall energy use can have a major impact on the environment. Today, fossil fuels are the primary source of energy for most individuals, businesses, and industries. However, their combustion is a significant source of greenhouse gas emissions and may contribute to other environmental problems such as acid rain. Renewable energy sources such as solar, wind, and hydropower may offer ways to reduce carbon emissions but come with potential environmental impacts of their own.

8.Weather events

Extreme weather events—such as hurricanes, floods, wildfires, droughts, and snowstorms—are becoming more frequent and severe due to climate change. These events pose a threat to both the environment and human populations, and can cause significant damage to infrastructure, homes and ways of life.

Rising temperatures and rising sea levels, among other factors, contribute to the increase in extreme weather conditions.According to the World Meteorological Organization, extreme weather and climate events were responsible for 2 million deaths and USD 4.3 trillion in economic losses between 1970 and 2021.[7]
Mixture of Experts | 23 January, episode 91

How are people and organizations addressing environmental issues?

Individuals, governments, organizations, and communities are attempting to address environmental issues in various ways:
Research and education

Through scientific research and development, people are learning more about the causes of environmental issues and ways to limit or repair their negative consequences. New technologies, including advances in renewable energy sources and energy-efficient operations, can help reduce carbon emissions.

Ecology and environmental science can offer a better understanding of how human activities affect Earth in complex ways. Public education efforts spread knowledge about the harmful impact of some behaviors and practices on the environment and can drive interest in environmental protection and sustainable development.
International cooperation

Collaborative international efforts are attempting to find and implement solutions that mitigate environmental problems. Environment, health and safety (EHS) organizations focus on safeguarding the environment from pollution and degradation. They play a role in researching and advising on environmental management and protection measures to reduce the negative impact of human activities, emissions, and hazardous materials on ecosystems.

Treaties, such as the Paris Agreement, are aimed at addressing climate change on an international scale. Sharing resources and best practices between countries can also lead to more effective environmental protection efforts.
Corporate responsibility

More companies are adopting corporate social responsibility policies and operating with environmental challenges in mind. They seek to increase sustainability in business to minimize any contribution to environmental change. Their efforts might include reducing emissions and their carbon footprint, by using less water and energy, and minimizing waste and improving waste management.

Transparency in reporting their environmental impact, by using environmental, social, and governance (ESG) reporting frameworks. And as mandated by directives like Europe's Corporate Sustainability Reporting Directive (CSRD), helps hold companies accountable and allows consumers to make informed choices.

Many are following the recommendations set forth by the Task Force on Climate-related Financial Disclosures, or TCFD, to inform investors, shareholders, and the public of their climate-related financial risks. Efforts to address environmental impact might help businesses find cost-saving measures and build trust and loyalty from consumers and investors.

Economic incentives

Some governments use economic incentives, such as taxes and subsidies to address environmental problems and foster environmentally friendly practices. For example, some used carbon taxes in an attempt to deter pollution-causing activities or used subsidies to help make electric cars or solar panels more affordable. Investments in sustainable infrastructure, such as public transportation and energy-efficient buildings, can also provide long-term economic and social benefits while reducing environmental impact.

Individual action

Some people are making lifestyle choices with the goal of reducing their environmental impact at an individual level. These choices may include reducing energy use, recycling, choosing sustainable products, or reducing water waste. Individuals can also advocate for environmental issues and support policies and businesses that prioritize sustainability.

Environmental Awareness Movements in India

Many environmental movements have emerged in India, especially after the 1970s. These movements have grown out of a series of independent responses to local issues in different places at different times.

Some of the best known environmental movements in India have been briefly described below:

1.The Silent Valley Movement

- The silent valley is located in the Palghat district of Kerala.
- It is surrounded by different hills of the State.
- The idea of a dam on the river Kunthipuzha in this hill system was conceived by the British in 1929.
- The technical feasibility survey was carried out in 1958 and the project was sanctioned by the Planning Commission of the Government of India in 1973.
- In 1978, the movement against the project from all corners was raised from all sections of the population.
- The movement was first initiated by the local people and was subsequently taken over by the **Kerala Sastra Sahitya Parishad (KSSP).**
- Many environmental groups like the Narmada Bachao Andolan (NBA), Bombay Natural History Society (BNHS) and Silent Valley Action Forum participated in the campaign.

2.Chipko Movement

- Chipko Movement started on April 24, 1973, at Mandal of Chamoli district of Gharwal division of Uttarakhand.
- The Chipko is one of the world-known environmental movements in India.
- The movement was raised out of ecological destabilisation in the hills.
- The fall in the productivity of the forest produces forced the hill dwellers to depend on the market, which became a central concern for the inhabitants.
- Forest resource exploitation was considered the reason behind natural calamities like floods, and landslides.
- On March 27 the decision was taken to ‘Chipko” that is ‘to hug’ the trees that were threatened by the axe and thus the chipko Andolan (movement) was born.
- This form of protest was instrumental in driving away the private companies from felling the ash trees.

3.Bishnoi Movement

- This movement was led by Amrita Devi, in which around 363 people sacrificed their lives for the protection of their forests.
- This movement was the first of its kind to have developed the strategy of hugging or embracing the trees for their protection spontaneously.

4.Appiko Movement

- It is a movement inspired by the Chipko movement by the villagers of Western Ghats,
- In the Uttar Kannada region of Karnataka, the villagers of Western Ghats started the Appiko Chalewali movement during the month of September – November 1983.
- Here, the destruction of forest was caused due to commercial felling of trees for timber extraction.
- Natural forests of the region were felled by the contractors, which resulted in soil erosion and drying up of perennial water resources.
- In the Saklani village in Sirsi, the forest dwellers were prevented from collecting usufructs like twigs and dried branches and non-timber forest products for the purposes of fuelwood, fodder, honey etc. They were denied their customary rights to these products.
- In September 1983, women and youth of the region decided to launch a movement similar to Chipko, in South India.
- The agitation continued for 38 days, and this forced the state government to finally concede to their demands and withdraw the order for the felling of trees.

5.Narmada Bachao Andolan

- Narmada is one of the major rivers of the Indian Peninsula.
- The scope of the Sardar Sarovar project, a terminal reservoir on Narmada in Gujurat in fact is the main issue in the Narmada Water dispute.

6.Jungle Bachao Andolan

- Jungle Bachao Andolan began in the 1980s in the Singhbhum district of Bihar (presently in Jharkhand).

- It was a movement against the government's decision to grow commercial teak by replacing the natural Sal forests.
- The tribal community is the most affected by this decision as it disturbs the rights and livelihood of Adivasis of that region.
- This movement was widely spread in states like Bihar, Jharkhand and Odisha in various other forms.

1. **Institutions in the Environment Protection**

a. Educational Institutions

- **Schools and Universities**: Incorporating environmental education into curricula helps instill environmental values in students from an early age. Universities often conduct research on environmental issues, innovate sustainable technologies, and host awareness campaigns.
- **Environmental Clubs and Programs**: Many institutions form student-led clubs that engage in environmental activities such as tree planting, recycling programs, or clean-up drives.
- **Partnerships with NGOs**: Institutions often partner with non-governmental organizations to run large-scale environmental initiatives like conservation projects, reforestation efforts, or renewable energy adoption.

b. Governmental Institutions

- **Environmental Policies and Regulations**: Governments create regulations like pollution control laws, wildlife conservation acts, and emission standards to safeguard the environment.
- **Conservation Programs**: National parks, wildlife reserves, and protected areas are managed to conserve biodiversity.
- **Funding for Green Projects**: Public institutions offer grants and subsidies for renewable energy projects, clean technologies, and other eco-friendly initiatives.

c. Corporate Institutions

- **Sustainability Practices**: Companies are adopting corporate social responsibility (CSR) by integrating sustainable practices into their business operations (e.g., reducing carbon footprints, sustainable sourcing).

- **Green Certifications**: Businesses seek certifications such as LEED (Leadership in Energy and Environmental Design) for eco-friendly construction or energy-efficient practices.
- **Innovation for Green Technology**: Many corporations invest in research and development for green technologies like electric vehicles, renewable energy systems, and waste management solutions.

2. People in the Environment Protection

a. Role of Individuals

- **Personal Lifestyle Choices**: People can adopt eco-friendly habits such as using energy-efficient appliances, reducing water consumption, recycling, and using public transportation.
- **Advocacy and Activism**: Individuals can participate in or lead campaigns to raise awareness about environmental issues (e.g., climate change marches, tree-planting events).
- **Community Engagement**: Engaging with local communities to create green spaces, start community gardens, and organize clean-up events helps in spreading environmental awareness.
- **Consumer Power**: By choosing to support eco-friendly products, individuals can influence companies to adopt greener practices.

b. Environmental Leaders and Influencers

- People like **Greta Thunberg** (climate change activism), **David Attenborough** (nature documentaries), and **Wangari Maathai** (Green Belt Movement) have played major roles in shaping public awareness and action on environmental issues.
- **Local Leaders**: Many communities have local environmental heroes who work tirelessly on small-scale projects like reforestation, waste management, or marine conservation.

c. Collective Action

- **Grassroots Movements**: Local and global movements such as Fridays for Future or Earth Hour have united people in their call for environmental justice and action.
- **Social Media Campaigns**: Social media is a powerful tool for raising awareness and mobilizing people on issues such as plastic pollution, climate change, and wildlife conservation.

CHAPTER-2 Concept of Sustainability and Sustainable Development

Sustainability

Definition: Sustainability refers to the ability to maintain or support a process or practice over the long term without depleting resources or causing severe ecological damage. It involves balancing environmental, economic, and social needs to ensure that future generations can also meet their needs.

Key Principles:

1. **Environmental Protection**: Ensuring that natural resources are used efficiently and conserved to prevent environmental degradation. This includes reducing pollution, protecting ecosystems, and managing resources responsibly.
2. **Economic Viability**: Promoting economic activities that are both profitable and sustainable in the long term. This involves supporting industries and practices that contribute to economic stability while also considering their environmental impact.
3. **Social Equity**: Ensuring fair access to resources and opportunities for all people. This includes addressing social inequalities, supporting community well-being, and fostering inclusive decision-making processes.

Three Pillars of Sustainability:

1. **Environmental**: Focuses on minimizing ecological impact and ensuring natural systems can thrive.
2. **Economic**: Ensures that economic activities contribute to long-term prosperity without causing environmental harm.
3. **Social**: Addresses issues of equity, community development, and social justice.

Sustainable Development

Definition: Sustainable development is a broader concept that encompasses the idea of sustainability but focuses specifically on achieving economic growth and development in a manner that is sustainable over the long term. It aims to meet the needs of the present without compromising the ability of future generations to meet their own needs.

Key Objectives:

1. **Integration**: Combining economic, social, and environmental goals in development strategies to ensure that progress in one area does not undermine the others.
2. **Long-term Perspective**: Planning and acting with future generations in mind, ensuring that current practices do not deplete resources or degrade ecosystems.
3. **Inclusive Growth**: Ensuring that development benefits are widely distributed and that all segments of society have access to opportunities and resources.

Achieving Sustainable Development

Sustainable development can be achieved if we follow the following points:

- It can be achieved by restricting human activities.
- Technological development should be input effective and not input utilising.
- The rate of consumption should not surpass the rate of salvation.
- For renewable resources, the rate of consumption should not surpass the rate of production of renewable substitutes.
- All types of pollution should be minimised.
- It can be achieved by sensible use of natural resources.

Ways to achieve Sustainable Development

1. Population Control Measures: It includes limiting the population to a level that is within the carrying capacity of the ecosystem. Non-renewable resources are limited and diminishing rapidly. The rate of depletion of non-renewable resources should not surpass the rate of production of renewable resources.
2. Use of the Input Efficient Technology: Technological advancement should be input efficient rather than input consuming. In other words, more should be produced with a particular amount of inputs, as it decreases resource exploitation.
3. Restrict Use of Renewable Resources: The extraction of renewable resources should be done on a sustainable basis. Simply put, the rate of extraction should not exceed the rate of regeneration.
4. Control Pollution: It means limiting pollution emissions to a level that is within the absorption capacity of the environment.
5. Substitute Non-renewable Resources with Renewable Resources: Depletion of non-renewable resources has hindered the flow of services over time. In order to maintain this flow, it is essential to develop renewable substitutes. Simply put, the

rate of depletion of non-renewable resources should not be more than the rate of creation of renewable substitutes.
Besides the above-mentioned ways to achieve sustainable development, in 2015, 17 Sustainable Development Goals were formulated by the UN which they intend to achieve by the year 2030.

Strategies for Sustainable Development

1. Use of Non-conventional sources for energy: India relies heavily on thermal and hydroelectric plants to satisfy its power demands. Thermal power plants release large amounts of carbon dioxide, a greenhouse gas, as well as fly ash, which harms water, land, and the environment. Forests are flooded by power plants, which also obstruct the water cycle. Non-conventional sources of energy, such as wind power and solar rays are cleaner and greener methods for producing electricity that can successfully substitute thermal and hydropower without harming the environment. Even though both of these methods include high initial costs, the benefits gained from them can easily absorb the high cost.
2. Establishing/installing Mini-hydel Plants: Streams can be found almost anywhere in mountainous areas. The majority of such streams are perpetual, perennial, or flow all year around Mini-hydel plants which are set up to use the energy of such streams to propel tiny turbines, which generate electricity/power. Such power plants are environmentally friendly because they do not alter land-use patterns or interfere with the natural flow of water while producing enough electricity to meet local demand.
3. Cleaner fuel use in both rural and urban regions: Households in rural areas generally rely on wood, waste from animals, and other biomass as fuel. This consumption of fuel has a number of negative effects, including deforestation, a decrease in green cover, waste of cattle dung, air pollution, etc. To deal with this issue, LPG and gobar gas are being promoted. Subsidised LPG is available, and gobar gas plants are promoted through low-interest financing and subsidies. These are environmentally friendly and cleaner fuels that contribute significantly to pollution reduction.
4. Utilisation of bio-compost: After the green revolution, Indian farmers shifted to the use of chemical fertilisers while ignoring the use of compost to improve agricultural output and production. A rise in the use of chemical fertilisers, or excessive use of chemical fertilisers, has not only harmed land fertility/productivity but has also contaminated water sources, including groundwater. Due to the increased demand for organic food, farmers have now become aware of the use of compost and have started to use compost made from organic wastes, which maintains and improves soil productivity and does not contaminate food. Cattle are

maintained in certain areas of the country only for the production of dung, which is an essential fertiliser and soil conditioner.

5. Biopest control: With the advent of the green revolution, the use of chemical pesticides for higher yields has increased, which not only contaminates food items but also pollutes soil and water bodies, including groundwater.

To step up and take on the challenge:

- The use of plant-based pesticides is being promoted. **For example,** Neem-based pesticides are environmentally friendly and have no adverse effects.
- Measures are being taken to improve public knowledge about the use of different animals and birds (such as snakes, lizards, owls, and peacocks) in controlling pests.
- The practice of mix cropping of producing various products on the same ground in consecutive years has also helped the farmers in sustainable development.

6. Traditional practices and knowledge: Traditionally, Indian people have been connected to their environment. All agricultural, healthcare, housing, and transportation practices were environmentally friendly. However, we have drifted/gone far away from the traditional system as a result of modernisation. It has caused damage to the ecosystem and our rural heritage.

7. Awareness among individuals: Environmental degradation is a threat to India's ecosystem that can only be solved with people's participation. People do their best to safeguard assets for future generations at the micro-level/family level, but at the macro-level/national level, they do not consider the protection of natural wealth/resources for future generations.

Examples of Sustainable Development

- Wind energy
- Solar energy
- Crop rotation
- Sustainable construction
- Efficient water fixtures
- Green space
- Sustainable forestry

Sustainable Development Goals (SDGs): In 2015, the United Nations adopted the 2030 Agenda for Sustainable Development, which includes 17 Sustainable Development Goals (SDGs) designed to address global challenges. These goals

cover a range of issues including poverty, inequality, climate change, environmental degradation, peace, and justice.

Examples of SDGs:

- **No Poverty**: Eradicate poverty in all its forms everywhere.
- **Quality Education**: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
- **Clean Water and Sanitation**: Ensure availability and sustainable management of water and sanitation for all.
- **Climate Action**: Take urgent action to combat climate change and its impacts.

Challenges and Considerations

1. **Resource Management**: Balancing the use of natural resources with their renewal and regeneration to prevent depletion and maintain ecosystem health.
2. **Climate Change**: Addressing the impacts of climate change through mitigation (reducing emissions) and adaptation strategies.
3. **Cultural and Social Dynamics**: Ensuring that development respects cultural diversity and promotes social cohesion.
4. **Economic Disparities**: Tackling inequalities in wealth and opportunities to ensure that development benefits are equitably shared.

What is Environmental Crisis?

Environmental crisis refers to a situation when an environment fails to perform its vital function of life sustenance. The environment becomes suitable as soon as the following happens:

1. Resource extraction remains below the rate of resource generation.
2. Generation of waste remains within the absorption capacity of the environment.

Reasons for Environmental Crisis

(1) Population explosion

- The high rate of growth of population adversely affects the environment.
- It increases the demand for environmental resources, but their supply is limited.
- This results in overuse and misuse of resources.

(2) Rise in economic activity

- The rise in economic growth results in affluent consumption and production of goods and services.
- It generates wastes that are beyond the absorptive capacity of the environment.

(3) Rapid industrialization

- Rapid industrialization has led to deforestation, and depletion of natural resources.
- It leads to contamination of water due to the accumulation of increasing quantity of toxic substances and industrial wastes in the water bodies.

(4) Urbanization

- A large migration of population from rural to urban areas results in the fast growth of slum areas.
- It leads to the excess burden on the existing infrastructural activities.

(5) Deforestation

- Deforestation refers to cutting down of trees, clearing forest, etc.
- It adversely affects the environment and causes other problems.

(6) Increased use of insecticides, pesticides, and chemical fertilizers

- Farmers and workers suffer health problems due to the increased use of poisonous insecticides, pesticides, and chemical fertilizers.
- The crop generated also contains chemical elements in it.

Chapter-3

Rio Earth Summit 1992, Background, Objectives, Features, Importance

Rio Earth Summit 1992 marked a milestone in global environmental diplomacy, introducing sustainable development, Agenda 21, UNFCCC, CBD, and key principles.

The Rio Earth Summit 1992 also known as the United Nations Conference on Environment and Development, marked a turning point in the history of international environmental diplomacy. In this conference, the global community came together not just to debate about environmental issues in isolation but to situate them within a larger framework of human development, economic growth and social equity. Also known as the Earth Summit, the conference laid the foundation for integrating environmental concerns into the developmental policies of nations and gave shape to the idea of sustainable development. In this article, we are going to cover Rio Earth Summit 1992, background, objectives and features.

Rio Earth Summit 1992

The Rio Summit 1992, or UNCED, was held in Rio de Janeiro, Brazil, from 3 to 14 June 1992. It was one of the largest gatherings of heads of state and government representatives at that time, involving leaders and officials from 178 countries, as well as representatives from international organisations and thousands of NGOs.

The Secretary-General of the Conference was Maurice Strong, who had earlier played a key role in the Stockholm Conference of 1972. He famously called Rio a "historic moment for humanity," as it signified the global recognition that economic development and environmental protection are inseparable.

Unlike earlier summits that focused narrowly on environmental concerns, the Earth Summit adopted a broad and holistic approach, recognising that development, poverty eradication, and environmental protection must go hand in hand.

Rio Earth Summit 1992 Background

The history of the Rio Summit can be traced back to the 1972 United Nations Conference on the Human Environment held in Stockholm, Sweden. That conference was the first major global attempt to place environmental concerns on the international agenda. While Stockholm raised awareness, it did not lead to substantial binding commitments.

By the late 1980s, growing challenges such as climate change, ozone depletion, deforestation, desertification, and biodiversity loss pushed the world towards recognising the urgent need for collective action. The publication of the Brundtland Commission Report, "Our Common Future" (1987), which introduced the concept of sustainable development,defined as "development that meets the needs of the

present without compromising the ability of future generations to meet their own needs" further laid the intellectual groundwork for Rio.

Thus, the Rio Summit of 1992 was conceived as a platform to address the linkages between environment and development, to reconcile economic aspirations with ecological sustainability, and to set the tone for global environmental governance in the 21st century.

Rio Earth Summit 1992 Objectives

The Rio Earth Summit 1992 was not just about a a single issue but sought to provide a comprehensive framework for global cooperation. Its primary objectives included:

Addressing urgent environmental issues such as climate change, deforestation, and biodiversity loss.

Promoting sustainable development as the guiding framework for future policies.

Drafting international agreements to tackle global environmental challenges.

Establishing **Rio Earth Summit 1992 Features**

norms and principles for integrating environmental concerns into development planning.

Strengthening the role of NGOs and civil society in environmental governance.

The Rio Earth Summit 1992 has the following features:

Participation of 178 nations, making it one of the largest multilateral conferences of its time.

Adoption of legally binding treaties such as the UNFCCC and the Convention on Biological Diversity (CBD).

Introduction of soft law instruments like the Rio Declaration and Agenda 21, which though non-binding, influenced global policymaking.

Recognition of the principle of common but differentiated responsibilities (CBDR), acknowledging that while all states share responsibility for environmental protection, the obligations of developed and developing countries differ.

Active involvement of NGOs, civil society groups, and indigenous communities, reflecting a participatory approach to global governance.

Rio Earth Summit 1992 Outcomes

Governments across the world adopted various initiatives in order to support environment conservation. These included:

The Rio Declaration on Environment and Development

The Rio Declaration outlined 27 principles to guide sustainable development.

It reaffirmed the centrality of human beings in development (Principle 1).

It emphasised the principle of sovereign rights of states over natural resources while also stressing their responsibility to prevent environmental harm (Principle 2).

It introduced the principle of intergenerational equity, ensuring resources for future generations (Principle 3).

It highlighted the role of poverty eradication, women, youth, and indigenous knowledge in sustainable development.

It reinforced the principle of CBDR, placing a greater onus on developed countries to lead climate and sustainability efforts.

Agenda 21

Agenda 21 was an action plan for sustainable development in the 21st century. Though non-binding, it became a reference point for national and local policies.

It had four sections:

Social and Economic Dimensions

Conservation and Management of Resources

Strengthening the Role of Major Groups

Means of Implementation

It emphasised the role of local governments, encapsulated in the phrase “Think Globally, Act Locally.”

Rio Earth Summit 1992 Importance

The Rio Summit was a landmark because it:

Brought sustainable development into mainstream global discourse.

Established a balance between economic, social, and environmental objectives.

Strengthened the idea of global environmental governance.

Highlighted the importance of inclusive participation by NGOs and civil society.

Institutionalised the concept of CBDR, shaping future negotiations.

Environmental Conventions and Protocols

Various Environmental Conventions and Protocols have been taken up for protection of the environment. These conventions and protocols include:

International Environmental Conventions

These are legally binding agreements between states to address global or regional environmental challenges. They are essential because:

They harmonise laws and standards across borders.

They prevent a “race to the bottom” in environmental regulations.

They create a framework for collective action.

The Kyoto Protocol (1997)

The Kyoto Protocol was adopted under the UNFCCC framework.

Entered into force in 2005.

Imposed legally binding emission reduction targets on developed countries.

Phase I (2005–2012): aimed at a 5% reduction in emissions.

Phase II (2013–2020): aimed at an 18% reduction.

Laid the groundwork for later negotiations culminating in the Paris Agreement (2015).

Importance for India

For India, the Rio Summit was important due to several reasons. This included:

It provided a platform to articulate the developmental concerns of the Global South.

India supported the principle of CBDR, highlighting the historical responsibility of developed countries.

It enabled India to secure commitments for technology transfer and financial support from developed nations.

It influenced domestic policies such as the Environment Protection Act (1986) and later initiatives on climate change....

The Convention on Biological Diversity (CBD) (1993) and the Ramsar Convention on Wetlands (1971) are major international environmental treaties focused on conservation. The CBD provides a broad, global framework for biodiversity protection, while Ramsar serves as the lead implementation partner for wetland-specific ecosystems, with both sharing goals of sustainable use and habitat protection.

Convention on Biological Diversity (CBD)

- **Focus:** Comprehensive conservation of all biodiversity (ecosystems, species, genetic resources).
- **Objectives:** Conservation of biological diversity, sustainable use of its components, and fair/equitable sharing of benefits from genetic resources.

- **Approach:** Broad, holistic approach requiring national strategies for sustainable development.
- **Scope:** Global, covering all ecosystems.

Convention on Biological Diversity +3

Ramsar Convention on Wetlands

- **Focus:** Specifically protects wetlands (marshes, lakes, rivers, coastal areas).
- **Objectives:** "Wise use" of all wetlands and designating specific, internationally significant wetlands as Ramsar sites.
- **Approach:** Focused, site-based, and management-oriented (maintaining ecological character).
- **Scope:** Focused specifically on wetland ecosystems.

Key Connections and Collaboration

- **Joint Efforts:** The two conventions work together through joint work plans (e.g., 2011-2020) to ensure aligned, coordinated action for biodiversity.
- **Implementation:** The Ramsar Convention acts as the lead partner on wetlands for the CBD, supporting the Aichi Biodiversity Targets and subsequent global biodiversity frameworks.
- **Synergy:** Ramsar sites are a key tool for achieving the broader goals of the CBD.

Convention on Biological Diversity +3

Both treaties promote the conservation of natural resources, with Ramsar providing specialized, in-depth protection for wetlands within the broader framework established by the CBD.

- The Ramsar Convention on Wetlands - DCCEEW
  22 Jan 2026 — Water within these areas can be static or flowing; fresh, brackish or saline; and can include inland rivers and coastal or marine ...
- The Convention on Biological Diversity (CBD) and the Ramsar ...

The new Strategic Plan for Biodiversity (2011-2020) was adopted at the tenth meeting of the Conference of the Parties to the Conve...

Convention on Biological Diversity

- cbd-en.pdf - Convention on Biological Diversity
  OBJECTIVES. The objectives of this Convention, to be pursued in accordance with its relevant provisions, are the conservation of b...

  Convention on Biological Diversity

The Chemical Weapons Convention (CWC) at a Glance

The Chemical Weapons Convention (CWC) is a multilateral treaty that bans chemical weapons and requires their destruction within a specified period of time. The treaty is of unlimited duration and is far more comprehensive than the 1925 Geneva Protocol, which outlaws the use but not the possession of chemical weapons.

CWC negotiations started in 1980 in the UN Conference on Disarmament. The convention opened for signature on January 13, 1993, and entered into force on April 29, 1997.

The CWC is implemented by the Organization for the Prohibition of Chemical Weapons (OPCW), which is headquartered in The Hague with about 500 employees. The OPCW receives states-parties' declarations detailing chemical weapons-related activities or materials and relevant industrial activities. After receiving declarations, the OPCW inspects and monitors states-parties' facilities and activities that are relevant to the convention, to ensure compliance.

The CWC is open to all nations and currently has 193 states-parties. Israel has signed but has yet to ratify the convention. Three states have neither signed nor ratified the convention (Egypt, North Korea and South Sudan).

Prohibitions

The Chemical Weapons Convention prohibits:

- Developing, producing, acquiring, stockpiling, or retaining chemical weapons.
- The direct or indirect transfer of chemical weapons.
- Chemical weapons use or military preparation for use.
- Assisting, encouraging, or inducing other states to engage in CWC-prohibited activity.
- The use of riot control agents "as a method of warfare."

Declaration Requirements

The CWC requires states-parties to declare in writing to the OPCW their chemical weapons stockpiles, chemical weapons production facilities (CWPFs), relevant chemical industry facilities, and other weapons-related information. This must be done within 30 days of the convention's entry into force for each member state.

Chemical Weapons Stockpiles—States-parties must declare all chemical weapons stockpiles, which are broken down into three categories:

- Category 1: chemical weapons based on Schedule 1 chemicals, including VX and sarin. (See below for an explanation of "scheduled" chemicals.)
- Category 2: chemical weapons based on non-Schedule 1 chemicals, such as phosgene.
- Category 3: chemical weapons including unfilled munitions, devices and equipment designed specifically to employ chemical weapons.

Other weapons-related declarations states-parties must make include:

- Chemical weapons production facilities on their territories since January 1, 1946.
- Facilities (such as laboratories and test sites) designed, constructed, or used primarily for chemical weapons development since January 1, 1946.
- "Old" chemical weapons on their territories (chemical weapons manufactured before 1925 or those produced between 1925 and 1946 that have deteriorated to such an extent that they are no longer useable).
- "Abandoned" chemical weapons (abandoned by another state without consent on or after January 1, 1925).
- Plans for destroying weapons and facilities.
- All transfers or receipts of chemical weapons or chemical weapons-production equipment since January 1, 1946.
- All riot control agents in their possession.

Chemical Industry—The CWC requires states-parties to declare chemical industry facilities that produce or use chemicals of concern to the convention. These chemicals are grouped into "schedules," based on the risk they pose to the convention. A facility producing a Schedule 1 chemical is considered a Schedule 1 facility.

- Schedule 1 chemicals and precursors pose a "high risk" to the convention and are rarely used for peaceful purposes. States-parties may not retain these chemicals except in small quantities for research, medical, pharmaceutical, or defensive use. Many Schedule 1 chemicals have been stockpiled as chemical weapons.
- Schedule 2 chemicals are toxic chemicals that pose a "significant risk" to the convention and are precursors to the production of Schedule 1 or Schedule 2 chemicals. These chemicals are not produced in large quantities for commercial or other peaceful purposes.

- Schedule 3 chemicals are usually produced in large quantities for purposes not prohibited by the CWC but still pose a risk to the convention. Some of these chemicals have been stockpiled as chemical weapons.

The CWC also requires the declaration of facilities that produce certain nonscheduled chemicals.

United Nations Environment Programme (UNEP)

The UNEP is an international environmental authority engaged in establishing a global environmental agenda and promoting the efficient implementation of the environmental dimension of the United Nations Sustainable Development Programme.

- The rising pollution levels of the 1960s and 1970s prompted the international leadership to think about having laws and regulations in place for environmental concerns, along the likes of the International Labour Organisation (ILO), the world Health Organisation (WHO), etc.
- These concerns were addressed at the 1972 United Nations Conference on Human Environment (also known as the Stockholm Conference). The Conference led to the adoption of the Stockholm Declaration (Declaration on the Human Environment).
- The Conference also resulted in the formation of a management body for these concerns, which was later called the United Nations Environment Programme.
- Headquartered in Nairobi, the UNEP is headed by an Executive Director.

UNEP Objectives

The UNEP's stated mission is to offer leadership and promote partnership in caring for the environment through informing, inspiring and enabling countries and peoples to enhance their quality of life without compromising that of future generations.

The UNEP has a few focus areas, in which they prioritise work. They are:

1. Climate change
2. Ecosystem management
3. Disasters and conflicts
4. Environmental governance
5. Resource efficiency
6. Chemicals and waste
7. Environment under review

UNEP Functions

The major functions of the UNEP are discussed below:

- The UNEP engages in developing global conventions on the environment and related issues. It hosts the secretariats of various conventions such as:
    - Minamata Convention
    - United Nations Convention on Biological Diversity
    - Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)
    - Basel Convention
    - Stockholm Convention
    - Rotterdam Convention
    - Montreal Protocol
    - Vienna Convention
    - Convention on Migratory Species
    - Tehran Convention
    - Bamako Convention
    - Carpathian Convention
    - Climate & Clean Air Coalition (CCAC)

It promotes environmental science and related information.

It finances and implements developmental projects related to the environment.

It engages with national governments, NGOs, etc. in relation to environmental policy and implementation.

The UNEP also formulates treaties and guidelines in the domain of international trade in harmful chemicals, international waterways pollution and transboundary pollution of air.

It also awards and honours individuals as well as institutions that do stellar work in this field.

Major Programmes of the UNEP

- Earth Hour
- Clean up the World
- Billion Tree Campaign
- Seal the Deal
- Pain for the Planet
- Awareness and Preparedness for Emergencies at Local Level (APELL)
- TUNZA
- Faith for Earth

United Nations Environment Assembly (UNEA)

The UNEP's governing body is called the **United Nations Environment Assembly**, which is said to be the world's highest decision making body on the environment.

- It meets once in two years to establish priorities for international environmental policies and develop international environmental law.
- Formed in 2012, it is headed by a Bureau and its President.
- The Bureau comprises ten environment ministers of various countries who all hold two-year terms, based on geographical rotation.
- Currently, it has 193 member states (all UN member countries).

UNEP and India

India has had a close relationship with the UNEP since the programme's inception. There are many projects completed, as well as ongoing projects, of the UNEP in India.

- The UNEP's presence in India started in 2016 with an office at New Delhi.

- The nodal agency for India's interactions with the UNEP is the GOI's Ministry for Environment, Forests and Climate Change.
- The Permanent Representative of India to UNEP is India's High Commissioner for Kenya.
- India's annual financial contribution to the UNEP is to the tune of USD 100,000.
- The UNEP has recognised India's initiatives in the environment sector.
    - The UNEP awarded PM Narendra Modi with the 'Champions of the Earth' award along with French President Emmanuel Macron in the category 'policy leadership'.
    - This was in recognition of the, among others, the International Solar Alliance, initiated by India.

In 2019, India joined the Climate & Clean Air Coalition (CCAC), whose Secretariat is hosted by the UNEP.

India plans to work with CCAC nations on best practices and experiences for the effective implementation of the National Clean Air Programme (NCAP).

The full form of CITES is the Convention on International Trade in Endangered Species of Wild Fauna and Flora. CITES is an international agreement between governments with the objective of the preservation of the planet's plants and animals by ensuring that the international trade in their specimens does not threaten their survival. It was adopted in 1963 and entered into force in 1975.

CITES

The 19th Conference of the Parties to the Convention on International Trade in Endangered Species of Wild Fauna and Flora or CITES COP19 was held in Panama City (14th – 25th November 2022). India submitted three proposals What is CITES?

CITES was conceptualised in 1963 at a meeting of the (IUCN) International Union For Conservation Of Nature.

- It came into force in 1975 and consists of 183 member-countries till date that abide by CITES regulations by implementing legislation within their own borders to enforce those regulations.

- Located in Geneva, Switzerland, the CITES is administered by the United Nations under its UNEP (United Nations Environment Programme) Wing.
- The Convention of Parties to CITES is the supreme decision-making body of the Convention and comprises all its Parties.
- The last CoP (17th) was held at Johannesburg (South Africa), in 2016. India hosted CoP (3rd) in 1981.
- Although CITES is legally binding on the Parties, it does not take the place of national laws.
- Rather, it provides a framework to be respected by each Party, which has to adopt its own domestic legislation to ensure that CITES is implemented at the national level.

CITES Classification or CITES Appendix

CITES classifies plants and animals into three categories, based on how threatened they are.

- Roughly 5,600 species of animals and 30,000 species of plants are protected by CITES against over-exploitation through international trade.

They are listed under the three CITES Appendices that are mentioned below:

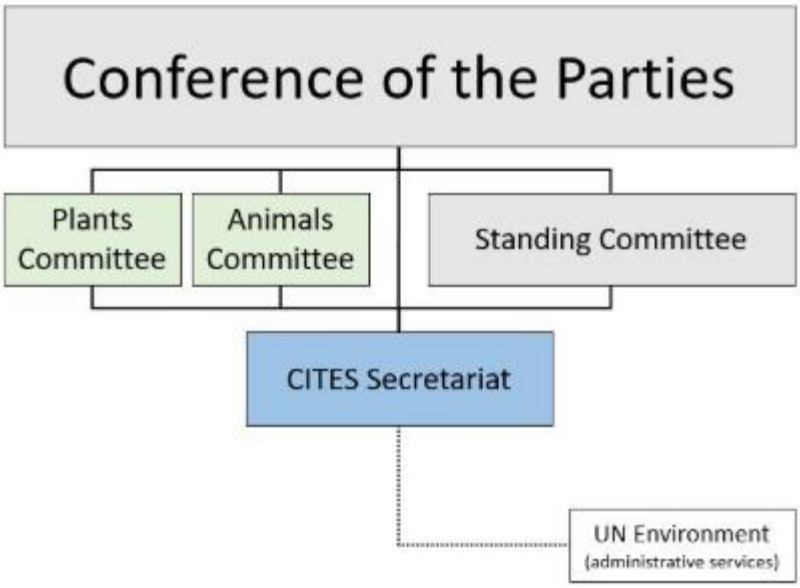
Appendix	Description	Examples of Species
CITES Appendix-I	• Species that are in **danger of extinction** • Commercial trade is prohibited. • Permits are required for import and export. • Trade permitted just for research only if the origin country ensures the trade won't harm the species' chance of survival.	• Asiatic lions and tigers (tiger skin trade). • Sea turtles, gorillas, lady slippers orchids (most species), etc. • Total 931 species on the list.

CITES Appendix-II	• Species that aren't facing imminent extinction but **need monitoring so that any trade doesn't become a threat**. • Trade permits obtained legally and only if the origin country ensures that its harvesting and trade won't harm the species' chance of survival.	• American Alligators (Alligator skin trade) • Paddlefish, Mahogany, corals, etc. • Total 34,419 species on the list.
CITES Appendix-III	• Species that are protected in at least one country. • Regulations for these species vary, but typically the country that requested the listing can issue export permits, and export from other countries requires a certificate of origin.	• Honeybadger (medicinal or bushmeat purpose) • Walruses, Map turtles, certain beetles, etc. • Total 147 species on the list.

- Species may be added to or removed from Appendix I and II, or moved between them, only by the Conference of the Parties.
- However, species may be added to or removed from Appendix III at any time and by any Party unilaterally.

Structure of CITES

The following image from the CITES official website (https://www.cites.org/) gives the structure:



CITES COP

The Conference of the Parties (COP) meet every two to three years. The **latest COP was CITES COP18** that took place in August 2019 in Geneva, Switzerland. **CITES COP3 took place in India** in 1981 in New Delhi.

CITES India

- India is a CITES Party since **1976**.
- Due to its extreme diversity, India is recognized all over the world for harbouring up to 7-8% of all the species recorded by CITES.
- Out of 34 global biodiversity hotspots in the world, India has 4 of them: Western Ghats, Sundaland, Himalayas and Indo-Burma region.
- As an active CITES Party, India prohibits the international trade of endangered wild species.
- India has placed several measures to control the threats from invasive alien species.
- This is done by regulating the trade by export certificates and import permits.

CITES COP19

India's proposal for transferring Leith's Softshell Turtle (*Nilssonia leithi*) and red-crowned roofed turtle (Batagur Kachua) from Appendix Ⅱ to Appendix Ⅰ has been adopted. The CITES Appendix Ⅰ listing will ensure that legal international trade of the species does not take place for commercial purposes. It will ensure the better survival of the species. Both turtle species are critically endangered and are illegally traded internationally as pets, food and traditional medicine.

- Nilssonia has been categorised as critically endangered by the IUCN. It is a large fresh-water, soft-shelled turtle which is endemic to peninsular India and inhabits rivers and reservoirs.
- India's proposal to add a gecko (*Cyrtodactylus jeyporensis*) to Appendix Ⅱ which would imply that international trade will require permits, has also been adopted. The Jeypore Indian gecko is endangered. The wild reptile species is

found in the Eastern Ghats and is known to be present in four locations including southern Odisha and northern Andhra Pradesh.

- India's proposals were among a total of 12 for greater protection of freshwater turtles and ultimately all 12 turtle proposals were adopted at COP19. It was highlighted that turtles are among the world's most threatened vertebrate groups.

India wants to re-list two otter species, star tortoise in CITES Appendix I, and Tokay gecko and Wedgefish in CITES Appendix